

Detailed Solution Key

Chapter 11: Work and Energy Worksheet

Exercise 1: The Volcanic Rock

1) Conservation of ME

Justification: Mechanical Energy is conserved because air resistance is neglected, meaning the only force doing work on the rock is its weight, which is a conservative force.

Formula: $ME_O = KE_O + GPE_O = \frac{1}{2}mV_0^2 + 0$

Numerical Substitution: $ME_O = 0.5 \times 20 \times (24)^2$

Final Answer: $ME_O = 10 \times 576 = \mathbf{5760\text{ J}}$.

2.1) GPE at highest point S

Formula: $GPE_S = m \cdot g \cdot H$

Numerical Substitution: $GPE_S = 20 \times 10 \times 3.4$

Final Answer: $GPE_S = \mathbf{680\text{ J}}$.

2.2) Speed V_S at highest point

Formula Setup: $ME_S = ME_O \Rightarrow KE_S + GPE_S = ME_O \Rightarrow \frac{1}{2}mV_S^2 + GPE_S = ME_O$

Numerical Substitution: $\frac{1}{2}(20)V_S^2 + 680 = 5760 \Rightarrow 10V_S^2 = 5080$

Final Answer: $V_S^2 = 508 \Rightarrow V_S = \sqrt{508} \approx \mathbf{22.54\text{ m/s}}$.

2.3) Angle of projection α

Formula Setup: $V_S = V_0 \cos \alpha \Rightarrow \cos \alpha = \frac{V_S}{V_0}$

Numerical Substitution: $\cos \alpha = \frac{22.54}{24} \approx 0.939$

Final Answer: $\alpha = \cos^{-1}(0.939) \approx \mathbf{20.1^\circ}$.

3) Speed V_A at ground via Work-Energy Theorem

Formula Setup: $\Delta KE = \sum W \Rightarrow \frac{1}{2}mV_A^2 - \frac{1}{2}mV_0^2 = W(\vec{W})$

Numerical Substitution: $\frac{1}{2}(20)V_A^2 - 5760 = m \cdot g \cdot h_{drop} = 20 \times 10 \times 36 = 7200$

Final Answer: $10V_A^2 = 7200 + 5760 = 12960 \Rightarrow V_A^2 = 1296 \Rightarrow V_A = \mathbf{36\text{ m/s}}$.

Exercise 2: Vertical Launch & Energy Graphs

Part A: Motion without air resistance

1.1) Absence of Air Resistance

Justification: The graph of Mechanical Energy (ME) as a function of altitude (z) is a horizontal straight line. This constant ME ($\Delta ME = 0$) proves that the work done by non-conservative forces (like air resistance) is zero.

1.2) Initial Speed V_0

Formula Setup: At launch ($z = 0$), the object is at the reference level, so $GPE_0 = 0$. Therefore, $ME = KE_0 = \frac{1}{2}mV_0^2$.

Numerical Substitution: Using the constant $ME = 10\text{ J}$ and $m = 2\text{ kg}$: $10 = \frac{1}{2}(2)V_0^2 \Rightarrow V_0^2 = 10$

Final Answer: $V_0 = \sqrt{10} \approx \mathbf{3.16\text{ m/s}}$.

- 2) **GPE Curve Description & Expression**
Curve Description: A straight line passing through the origin (0, 0) with a positive slope of 20.
Expression: $GPE = m \cdot g \cdot z = 2 \times 10 \times z = 20z$ (in Joules).
- 3) **Deducing the KE Expression**
Expression: Since $ME = KE + GPE \implies KE = ME - GPE = 10 - 20z$.
- 4) **Maximum Altitude Z_{max}**
Formula Setup: At the maximum height, $V = 0 \implies KE = 0$.
Numerical Substitution: $10 - 20z = 0 \implies 20z = 10$
Final Answer: $Z_{max} = 0.5 \text{ m}$.

Part B: Motion with air resistance

- 1) **Identifying the KE Curve**
Justification: The curve that starts at a maximum positive value at $z = 0$ and decreases represents the Kinetic Energy, because the object is launched with an initial speed that drops as it rises.
- 2) **Maximum Height Z_{max}**
Final Answer: Read from the horizontal z -axis where the Kinetic Energy curve reaches 0 J: $Z_{max} = 4 \text{ m}$.
- 3) **Thermal Energy Produced**
Formula Setup: $E_{th} = |\Delta ME| = ME_{initial} - ME_{final}$
Numerical Substitution: At $Z_{max} = 4 \text{ m}$, $KE = 0 \text{ J}$, so $ME_{final} = GPE_{final} = 8 \text{ J}$.
 $E_{th} = 10 - 8$
Final Answer: $E_{th} = 2 \text{ J}$.

Exercise 3: The Skier

- 1.1) **Mechanical Energy at point A**
Formula Setup: $ME_A = KE_A + GPE_A = \frac{1}{2}MV_A^2 + M \cdot g \cdot h_A$
Numerical Substitution: $ME_A = 0 + 60 \times 10 \times 3.5$
Final Answer: $ME_A = 2100 \text{ J}$.
- 1.2) **Proof of frictionless track AB**
Formula Setup: $ME_B = KE_B + GPE_B = \frac{1}{2}MV_B^2 + 0$
Numerical Substitution: $ME_B = 0.5 \times 60 \times (\sqrt{70})^2 = 30 \times 70 = 2100 \text{ J}$.
Justification: Since $ME_B = ME_A$, mechanical energy is conserved, meaning $W_{fric} = 0$, proving track AB is frictionless.
- 2.1) **Work done by external forces between B and C**
Work of Weight: $W(\vec{W}) = 0 \text{ J}$ ($\vec{W} \perp \vec{BC}$).
Work of Normal: $W(\vec{N}) = 0 \text{ J}$ ($\vec{N} \perp \vec{BC}$).
Work of Friction: $W(\vec{f}) = -f \cdot BC = -60 \times 4 = -240 \text{ J}$.
- 2.2) **Speed of the skier at C**
Formula Setup: $\Delta ME_{B \rightarrow C} = W(\vec{f}) \implies ME_C - ME_B = W(\vec{f})$
Numerical Substitution: $ME_C - 2100 = -240 \implies ME_C = 1860 \text{ J}$. $\frac{1}{2}(60)V_C^2 = 1860$
Final Answer: $30V_C^2 = 1860 \implies V_C = \sqrt{62} \approx 7.87 \text{ m/s}$.

2.3) Energy Conversion

Final Answer: Thermal Energy (or Heat).

3.1) Energies at C and D

3.1.1) ME_C : $ME_C = KE_C + 0 = 1860 \text{ J}$.

3.1.2) GPE_D : $GPE_D = 60 \times 10 \times 3.5 = 2100 \text{ J}$.

3.2) Deduction

Justification: Track CD is frictionless (ME conserved). To reach D, ME_C must be $\geq GPE_D$. Since $1860 < 2100$, the skier cannot reach D.

3.3) Height h_I of the highest point I

Formula Setup: At highest point, $V_I = 0$, so $ME_I = GPE_I$. Conservation: $GPE_I = ME_C$.

Numerical Substitution: $M \cdot g \cdot h_I = 1860 \implies 600 \cdot h_I = 1860$

Final Answer: $h_I = 3.1 \text{ m}$.

Exercise 4: The Track & Collision

1.1) Magnitude of friction force f on track AB

Formula Setup: $\Delta KE = W(\vec{W}) + W(\vec{f}) \implies \frac{1}{2}mV_B^2 - 0 = mgh_A - f \cdot AB$.

Numerical Substitution: $h_A = 4 \times 0.2 = 0.8 \text{ m}$. $\frac{1}{2}(2)(3)^2 = (2)(10)(0.8) - f(4) \implies 9 = 16 - 4f$.

Final Answer: $4f = 7 \implies f = 1.75 \text{ N}$.

1.2) Instantaneous Power at $V = 2 \text{ m/s}$

Friction: $P_f = -f \cdot V = -1.75 \times 2 = -3.5 \text{ W}$.

Normal: $P_N = N \cdot V \cos(90^\circ) = 0 \text{ W}$.

Weight: $P_W = mg \sin \alpha \cdot V = 2 \times 10 \times 0.2 \times 2 = 8.0 \text{ W}$.

1.3) Average Power of friction

Formula Setup: $P_{avg} = \frac{W(\vec{f})}{\Delta t} = \frac{-f \cdot AB}{\Delta t}$.

Numerical Substitution: $P_{avg} = \frac{-1.75 \times 4}{2.7} = \frac{-7}{2.7}$.

Final Answer: $P_{avg} \approx -2.59 \text{ W}$.

2) Maximum deviation angle β

Formula Setup: $ME_C = ME_D \implies \frac{1}{2}MV_0^2 = Mgl(1 - \cos \beta)$.

Numerical Substitution: $\frac{1}{2}(3)(1.6)^2 = 3 \times 10 \times 0.5(1 - \cos \beta) \implies 3.84 = 15(1 - \cos \beta)$.

Final Answer: $\cos \beta = 1 - 0.256 = 0.744 \implies \beta = \cos^{-1}(0.744) \approx 41.9^\circ$.

3.1) Mechanical Energy of (P-Earth) at D

Final Answer: $ME_D = ME_C = \frac{1}{2}(3)(1.6)^2 = 3.84 \text{ J}$.

3.2) Speed and altitude when $KE = GPE$

Formula Setup: $KE + GPE = 3.84 \implies 2GPE = 3.84 \implies GPE = 1.92 \text{ J}$ and $KE = 1.92 \text{ J}$.

Altitude: $Mgh = 1.92 \implies 30h = 1.92 \implies h = 0.064 \text{ m}$.

Speed: $\frac{1}{2}MV^2 = 1.92 \implies 1.5V^2 = 1.92 \implies V = 1.13 \text{ m/s}$.

3.3) Speed V_{ground} upon reaching the ground

Final Answer: $ME_{ground} = ME_D = 3.84 \text{ J} \implies \frac{1}{2}MV_{ground}^2 = 3.84 \implies V_{ground} = 1.6 \text{ m/s}$.

Exercise 5: Spring Launch

a) Relation among variables

Formula Setup: Mechanical Energy is conserved between launch (L) and point A. $ME_L = ME_A$.

Expression at Launch: $ME_L = EPE = \frac{1}{2}k(L_0 - L)^2$ (since $v = 0, z = 0$).

Expression at A: $ME_A = KE_A + GPE_A = \frac{1}{2}mv^2 + mgZ$.

Final Relation: $\frac{1}{2}k(L_0 - L)^2 = \frac{1}{2}mv^2 + mgZ$.

b) Calculate Spring Constant k

Formula Setup: At max altitude, $v = 0$, so $\frac{1}{2}k(L_0 - L)^2 = mgZ_{max} \Rightarrow k = \frac{2mgZ_{max}}{(L_0 - L)^2}$.

Numerical Substitution: Convert to standard units: $m = 0.6 \text{ kg}$, $Z_{max} = 0.2 \text{ m}$, $\Delta L = (0.25 - 0.20) = 0.05 \text{ m}$.

$$k = \frac{2 \times 0.6 \times 10 \times 0.2}{(0.05)^2}$$

Final Answer: $k = \frac{2.4}{0.0025} = 960 \text{ N/m}$.

c) Instantaneous Power of Weight at A

Formula Setup: $P_{\vec{W}} = \vec{W} \cdot \vec{v} = W \cdot v \cdot \cos(\theta)$, where θ is the angle between downward gravity and velocity.

Numerical Substitution: The angle between vertical gravity and the incline velocity is $(90^\circ + \alpha)$.

$$P_{\vec{W}} = -mg \cdot v \cdot \sin \alpha.$$

Final Answer: Since point A is defined generically with velocity v , the specific numerical value cannot be calculated without v or α . The algebraic expression is $-mgv \sin \alpha$.

Exercise 6: Inclined Plane & Energy Graph

1) Calculate Initial Speed V_0

Formula Setup: From the graph, at $x = 0$, $KE_0 = 1.0 \text{ J}$. $KE_0 = \frac{1}{2}mV_0^2$.

Numerical Substitution: $1.0 = \frac{1}{2}(0.2)V_0^2 \Rightarrow 1.0 = 0.1V_0^2$.

Final Answer: $V_0^2 = 10 \Rightarrow V_0 = \sqrt{10} \approx 3.16 \text{ m/s}$.

2) Speed at $x = 4 \text{ m}$

Formula Setup: From the graph, at $x = 4$, $KE_4 = 3.0 \text{ J}$. $KE_4 = \frac{1}{2}mV_4^2$.

Numerical Substitution: $3.0 = \frac{1}{2}(0.2)V_4^2 \Rightarrow 3.0 = 0.1V_4^2$.

Final Answer: $V_4^2 = 30 \Rightarrow V_4 = \sqrt{30} \approx 5.48 \text{ m/s}$.

3) Sum of Works $\Sigma W(\vec{F}_{ext})$

Formula Setup: Work-Kinetic Energy Theorem: $\Sigma W(\vec{F}_{ext}) = \Delta KE = KE_4 - KE_0$.

Numerical Substitution: $\Sigma W(\vec{F}_{ext}) = 3.0 \text{ J} - 1.0 \text{ J}$.

Final Answer: $\Sigma W(\vec{F}_{ext}) = 2.0 \text{ J}$.

4) Algebraic Expression & Friction Value

Formula Setup: The external forces doing work are Weight ($\vec{W}$) and Friction ($\vec{f}$).

$$\Sigma W = W(\vec{W}) + W(\vec{f}) = mg(x_A \sin \alpha) - f \cdot x_A.$$

Numerical Substitution: $2.0 = (0.2)(10)(4 \sin 30^\circ) - f(4) \Rightarrow 2.0 = 2(4 \times 0.5) - 4f \Rightarrow 2.0 = 4 - 4f$.

Final Answer: $4f = 2.0 \Rightarrow f = 0.5 \text{ N}$.

5) Instantaneous Power at $x = 4 \text{ m}$

Formula Setup: $P_{f_{inst}} = \vec{f} \cdot \vec{v}_4 = -f \cdot V_4$.

Numerical Substitution: $P_{f_{inst}} = -0.5 \times 5.48$.

Final Answer: $P_{f_{inst}} = -2.74 \text{ W}$.

6) Average Power of Friction

Formula Setup: $P_{avg} = \frac{W(\vec{f})}{\Delta t} = \frac{-\vec{f} \cdot \vec{x}_A}{\Delta t}$.

Numerical Substitution: $P_{avg} = \frac{-0.5 \times 4}{0.92} = \frac{-2.0}{0.92}$.

Final Answer: $P_{avg} \approx -2.17 \text{ W}$.